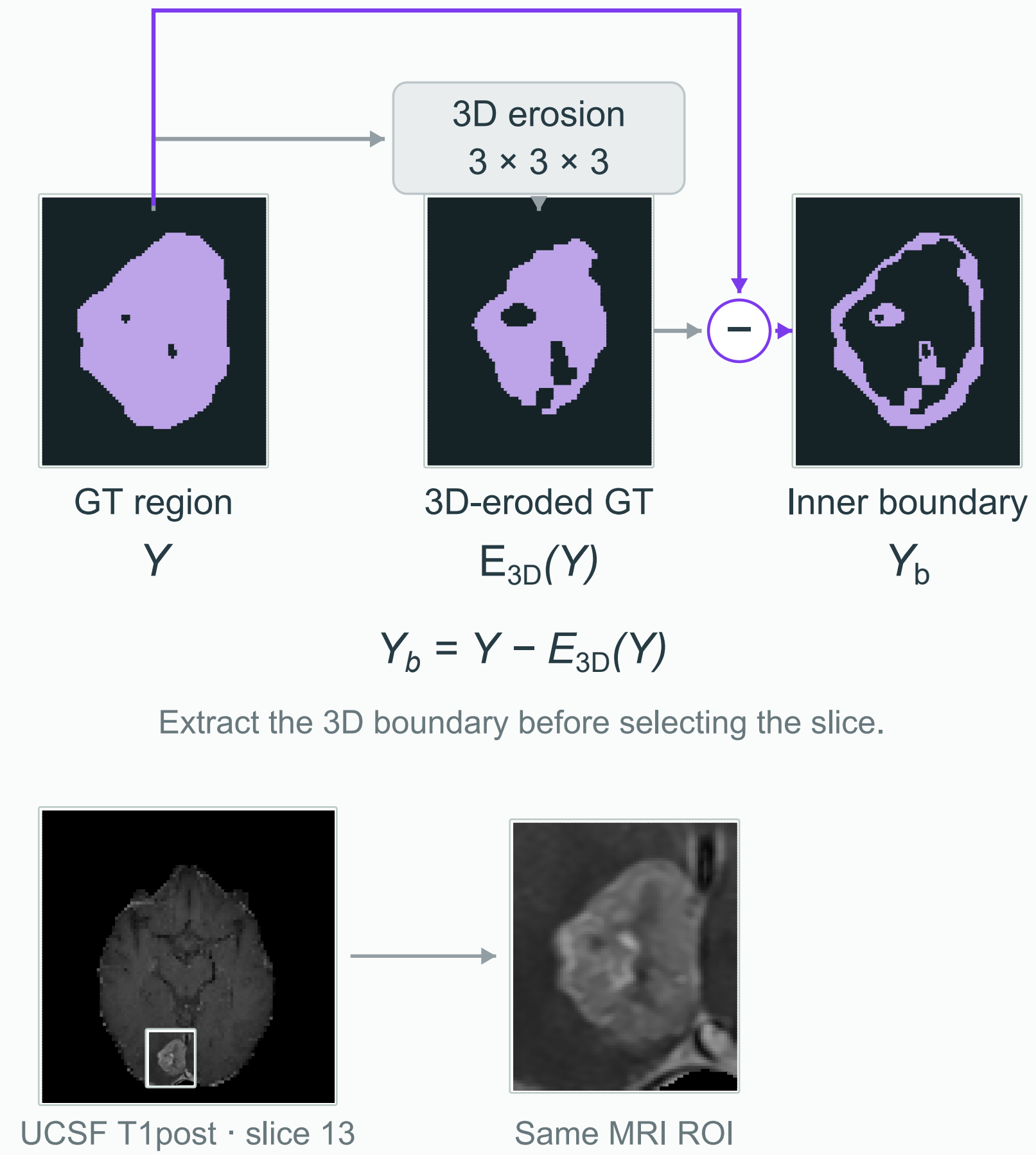


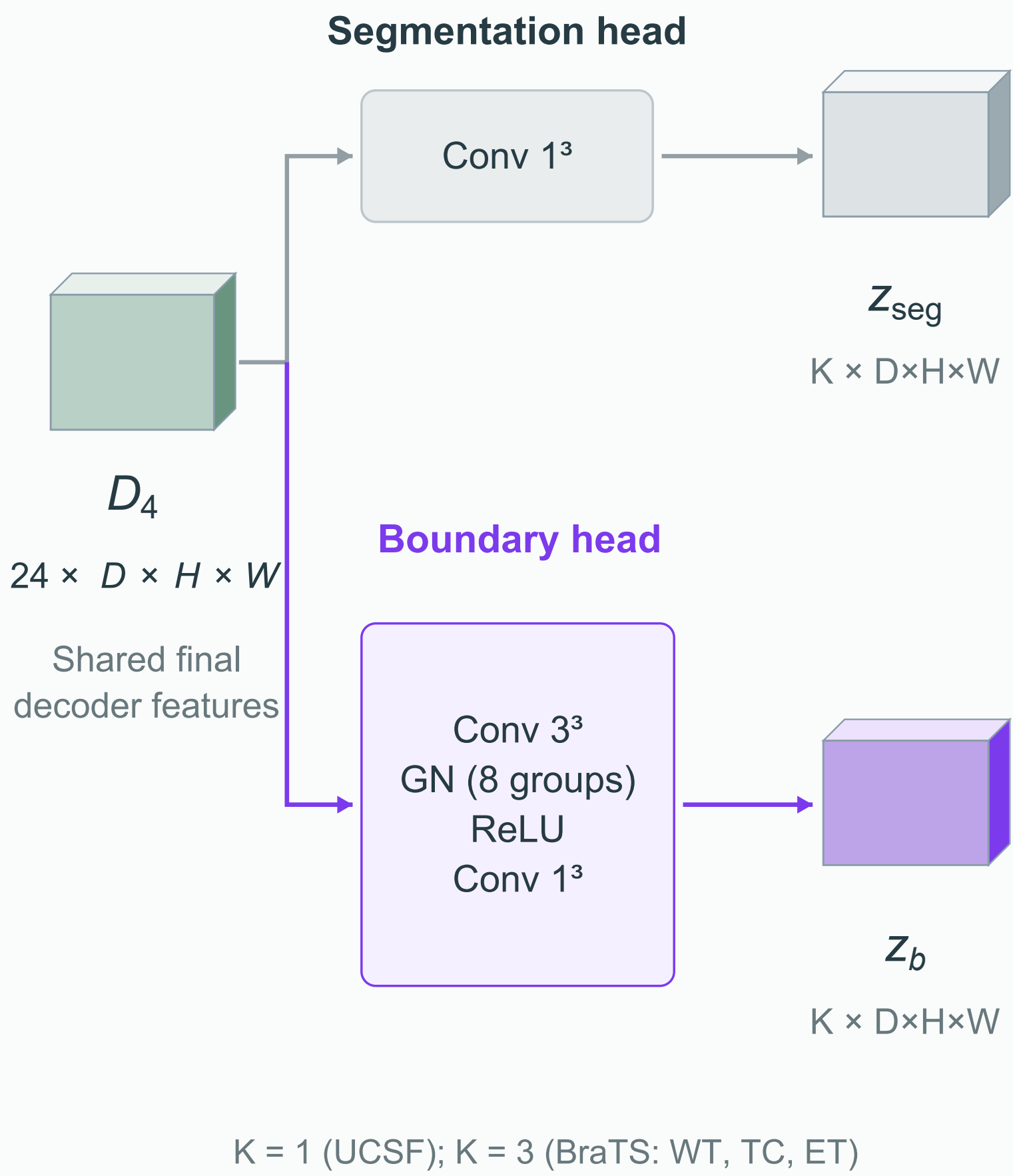
ABS: auxiliary boundary supervision

Real volumetric boundary labels · Shared decoder features · Joint segmentation and boundary objectives

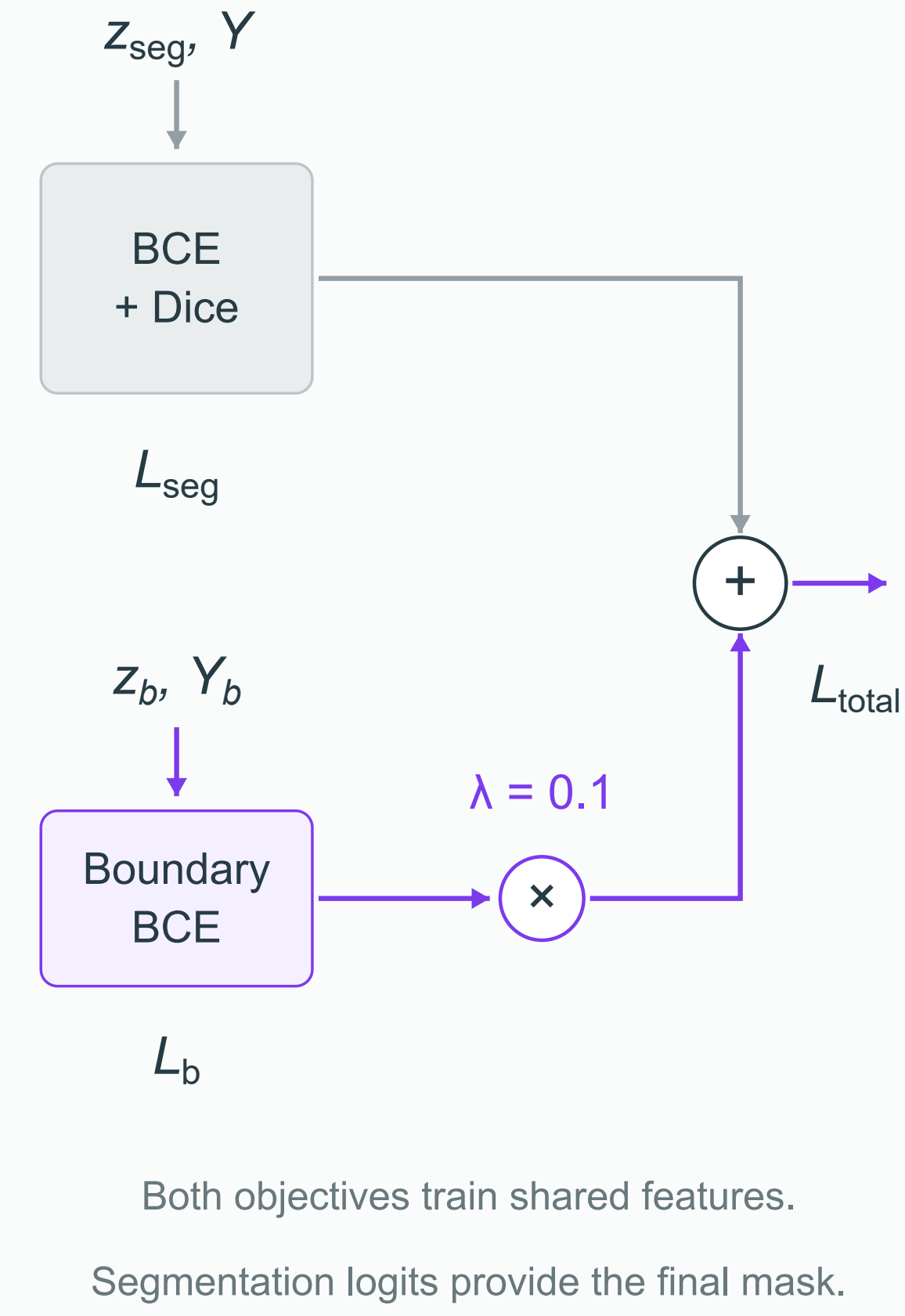
(a) Boundary target construction



(b) Dual prediction heads



(c) Joint supervision



$$L_{total} = BCE(z_{seg}, Y) + Dice(z_{seg}, Y) + 0.1 BCE(z_b, Y_b)$$

GN: GroupNorm · BCE: binary cross-entropy with logits · Dashed boxes: module panels · Purple: auxiliary boundary branch